

University of Illinois at Chicago

Department of Chemical Engineering

SAFETY QUIZ: CHE381- Chemical Engineering Laboratory I

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?
Awareness (of chemical hazards, appropriate safeguards, proper use, proper storage, hygiene, methods, and emergency procedures).
2. What two pieces of personal protective equipment must be worn at all times in the lab?
Safety glasses and safety shoes
3. What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?
Lab coats, gloves, splash aprons, face shields, and ear protection.
4. True or False: Fume hoods are generally used as storage areas for chemicals.
False
5. Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:
1000 ppm or 1000 mg/M3
6. What are the four routes of chemical entry into the body?
Inhalation, ingestion, injection, and eye or skin contact.
7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.
False

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8. What is the role of the Chemical Hygiene Plan?

To coordinate safety procedures so that every employee can be trained according to the details of the Plan.

9. What is the guideline for laboratory room airflow and fume hood airflow?

It is 20 linear feet per minute airflow for the laboratory room and 80-120 linear feet per minute for fume hoods

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

- Keeping aisles, stairs, and hallways clear of chemicals
- Properly labeling wastes and keeping them in proper containers.
- Working surfaces and floors should be cleaned

11. What is the proper method of transporting gas cylinders in the laboratory?

They should be transported using 1-cylinder, 4-wheel carts. The cap must be on the cylinder during transportation.

12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

False

13. What is an accepted method of vapor detection in the laboratory?

Draeger tubes

14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.

- Wearing appropriate eye protection
- Avoiding inhalation of chemicals
- Not eating or drinking in a laboratory

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15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.

They can be transported via an appropriate “bucket” type carrier (Nalgene Safety Bottle Carrier, Safety Bottle Carrier) which is shock resistant. The person transporting the chemical must be wearing safety glasses.

16. List the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory.

- Evacuate the area
 - Dial “0” and report type of emergency and location.
 - Operator will notify employees to evacuate, and will activate the Emergency Response Team
 - Exit the lab and deactivate power by pushing the emergency power button outside of lab
 - Evacuate the building
 - Assemble in front of the facility
 - Report to department secretary
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17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

A flammable chemical has a flashpoint below 100°F while a combustible chemical has a flashpoint between 100°F and 200°F.

18. List the Emergency Procedure for an individual chemical exposure in the laboratory.

- Dial “0” and report type of emergency and location
 - Operator calls Fire Department Paramedics and notifies Emergency Response Team
 - If there are no apparent dangers:
 - If a person has been splashed with a chemical, wash them with plenty of water for at least 15 minutes and remove contaminated clothing.
 - If a person has inhaled a chemical, get them to fresh air and apply artificial respiration if necessary.
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19. What is the definition of a “hazardous chemical”?

A chemical that may cause an acute or chronic health effect.

20. All laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?

The flow rate for safety showers should be of 30 gal/min and for eyewashes of 3 gal/min. They must be checked once a month.
